

Introduction to neural networks

Machine learning recall

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26 de agosto de 2026



- 1 Introduction to machine learning
- 2 Machine learning tasks
- 3 Types of machine learning

Machine Learning

A field of study that gives computers the ability to learn without being explicitly programmed ^a

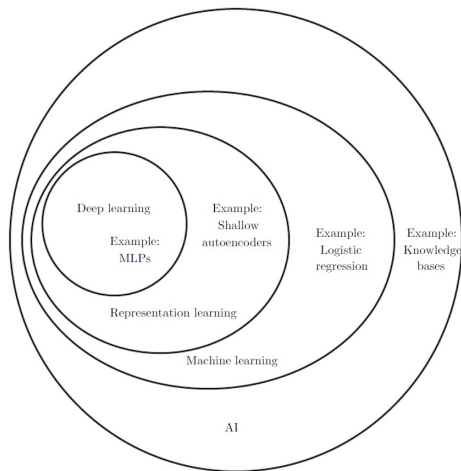
^aArthur Samuel, 1959

Learning

An agent is learning if it improves its performance on future tasks after making observations about the world. ^a

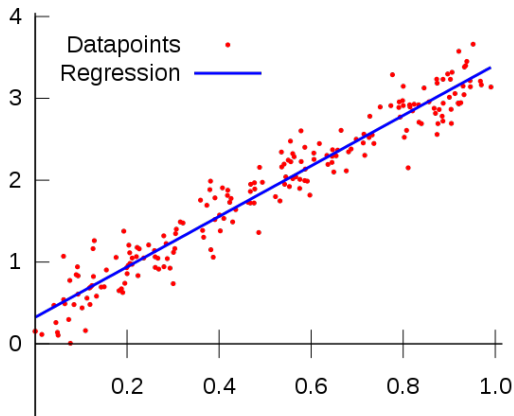
^aRussell and Norvig

Machine Learning in Artificial Intelligence ¹



¹Goodfellow, I., Bengio, Y., and Courville, A. (2016). Deep learning. MIT press.

- **Regression.** Finds a function that relates the input to a continuous codomain.



Machine Learning

Tasks

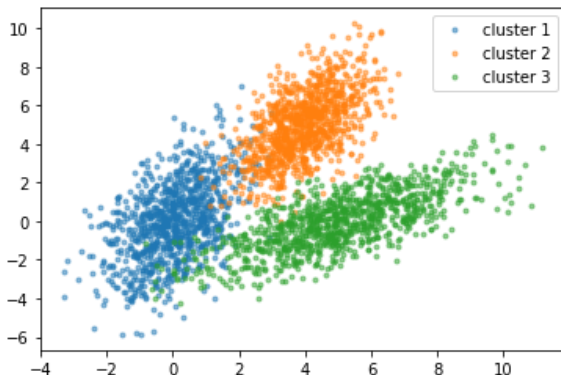
- **Classification.** Assigns a label to a given input. The set of classes is predefined.



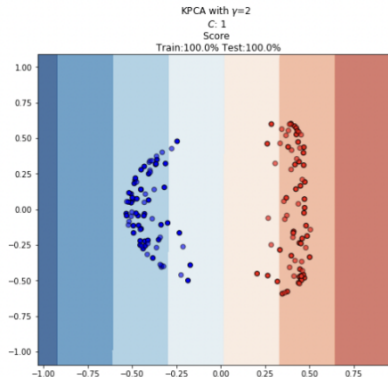
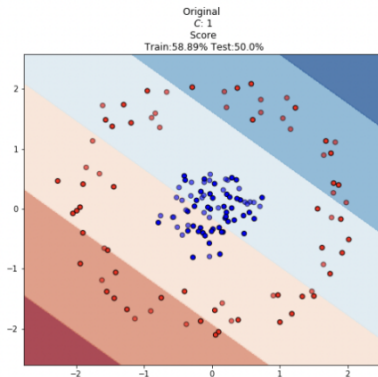
or



- **Clustering.** Automatically groups the inputs according to their features. The number of groups may be unknown.



- **Dimensionality reduction.** Describes an entity with less features.



Machine Learning

Tasks

- **Generative modeling.** To create new instances from a distribution.



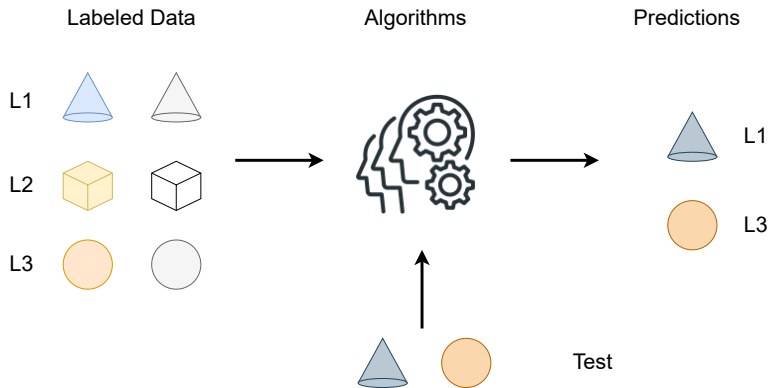
Machine Learning

Types

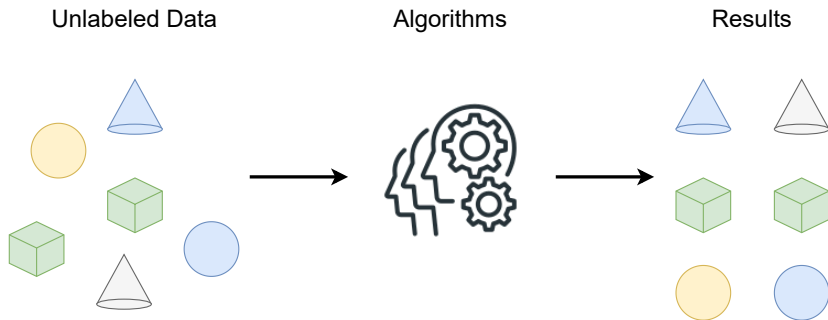
Depending on how machines learn:

- **Supervised Learning.** The machine learns from labeled data.
- **Un-supervised Learning.** The machine tries to discover the structure of input data without labels.
- **Reinforcement Learning.** The machine interacts with the environment improving its actions.

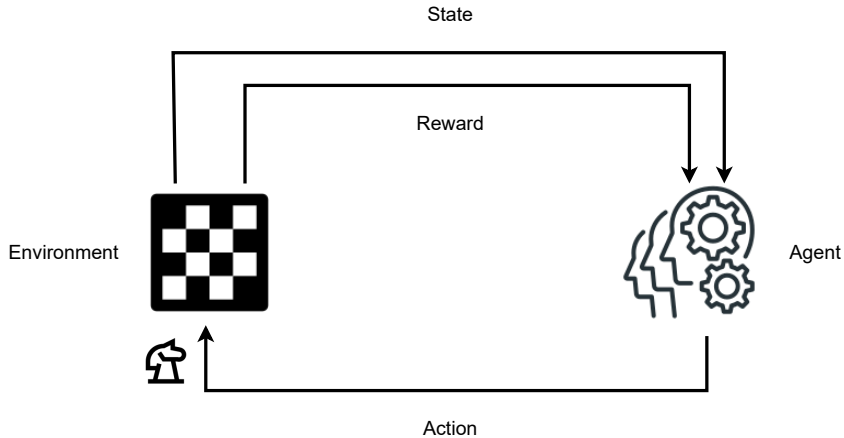
Supervised Learning



Unsupervised Learning



Reinforcement Learning



- Sections 18.1 and 18.2, *Learning from examples* chapter, Russell and Norvig, Artificial Intelligence: A modern approach.
- Section 5.1 *Learning Algorithms*. Goodfellow et al., Deep Learning, MIT Press.

- <https://www.topbots.com/ai-research-generative-adversarial-network-images/>
- <https://neptune.ai/blog/dimensionality-reduction>
- <https://www.kaggle.com/c/dogs-vs-cats>
- https://training.galaxyproject.org/archive/2022-08-01/topics/statistics/tutorials/clustering_machinelearning/tutorial.html
- Photo by charlesdeluvio on Unsplash, Photos: Julie Molliver Shane Rounce